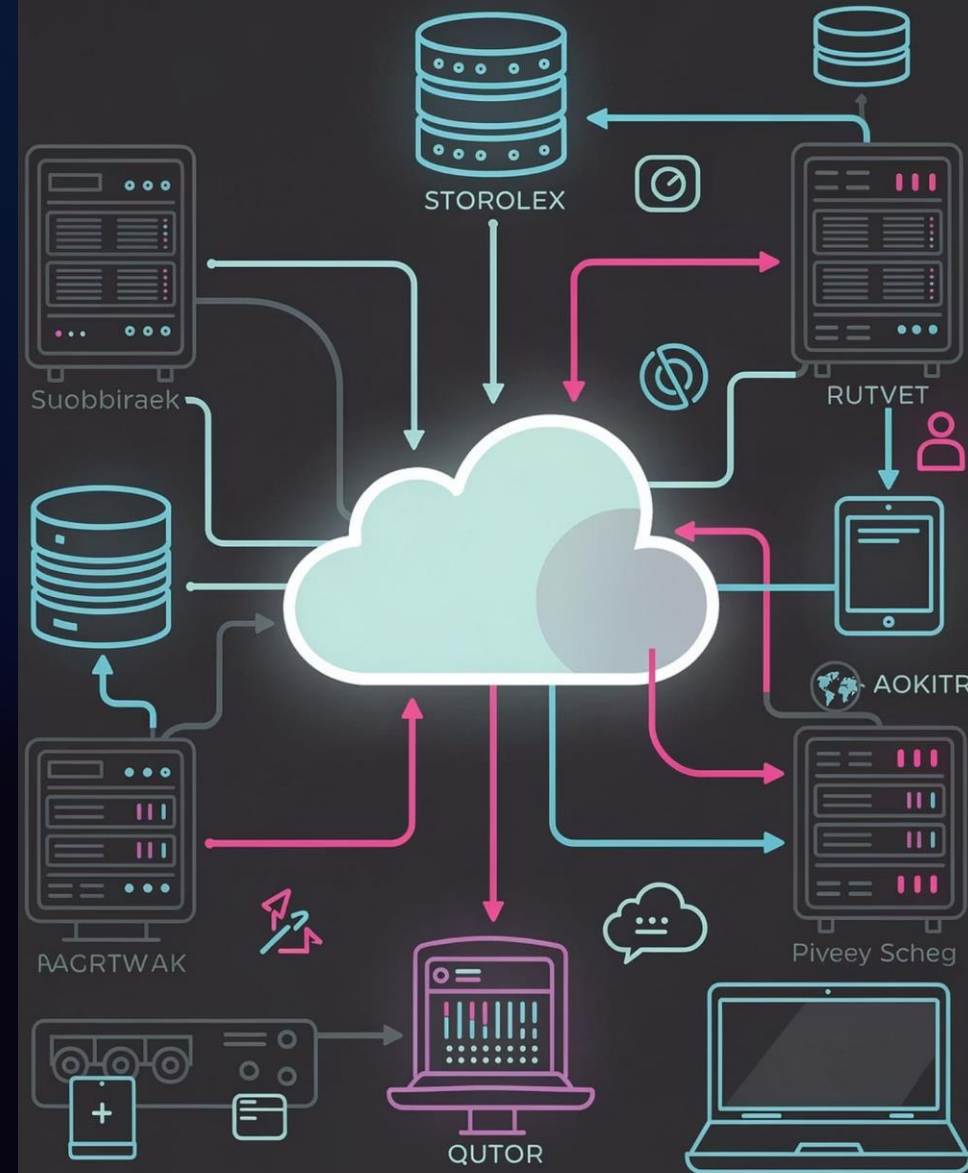


Day 8: Virtual Private Cloud (VPC) Networking Essentials

Welcome to one of the most important lessons in your cloud career toolkit. Today we're building the skills that employers pay real money for — knowing how to design, secure, and manage a cloud network from the ground up.

ITNW 1009 · DAY 8



Today's Agenda

What We're Covering Today

01

VPC Fundamentals

Your own isolated network inside the AWS Cloud

02

CIDR & Subnets

Carving up your IP address space into neighborhoods

03

Gateways & Route Tables

Controlling how traffic flows in and out

04

Firewalls: SGs & NACLs

Two layers of protection that work together

05

Real-World Case Study

A healthcare breach — and how to prevent it

Key Vocabulary

Terms You'll Own by End of Class

Don't let the acronyms intimidate you. Every one of these terms maps to something you already understand from everyday life. By the end of today, these will feel completely natural.

VPC

Virtual Private Cloud – your isolated network bubble inside AWS

CIDR

Your IP address range – the zip code for your cloud neighborhood

Subnet

Public or Private – distinct sections inside your VPC land

IGW / NAT

Gateways controlling what traffic can enter or exit

SG / NACL

Two firewall types – your instance door guard and subnet perimeter gate

Why This Lesson Matters for Your Career

In a traditional office environment, building a secure network meant ordering physical routers, running ethernet cables under floors, and mounting hardware onto racks. One wiring mistake and your database could be wide open to the internet.

The Cloud Difference

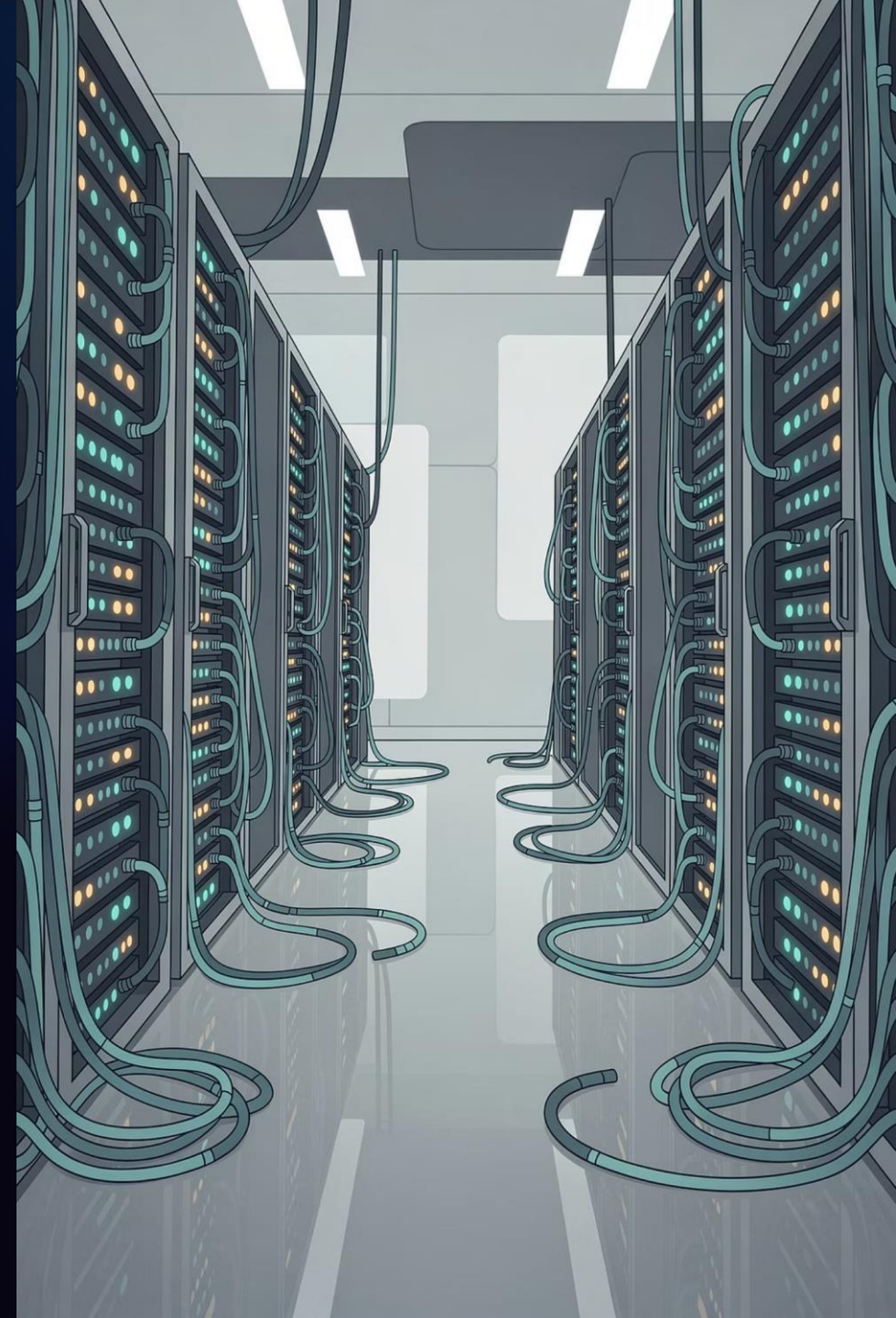
All of that physical infrastructure is now virtualized — built and managed through software, instantly, without touching a single cable.

Employer Demand

Network operations specialists and cloud engineers who understand VPC architecture are among the most sought-after professionals in IT today.

Your Competitive Edge

Mastering how to lock down a corporate cloud network is your bread and butter — it separates entry-level from job-ready.



Chapter 1

Introduction to AWS Virtual Private Cloud

Your own isolated data center — entirely in software

What Is an Amazon VPC?

The Core Concept

An Amazon VPC — Virtual Private Cloud — is your own completely isolated, private network environment hosted inside the AWS Cloud. Think of it as renting your own private, fenced-in plot of digital land inside a highly protected corporate business park.

Everything inside this perimeter belongs exclusively to your organization. No one can enter unless you explicitly build a gateway and open the door.

What You Control

- The **layout** of your network — how subnets are arranged
- The **IP address ranges** assigned to each section
- The **routing rules** that govern traffic flow
- The **security policies** applied at every layer

📌 Networks are no longer made of expensive physical switches and cables. In the cloud, it's all software-defined — and you're in charge.

The Corporate Office Park Analogy

Imagine a massive, high-security corporate business park enclosed inside an industrial zone. This physical structure directly mirrors how a VPC works.



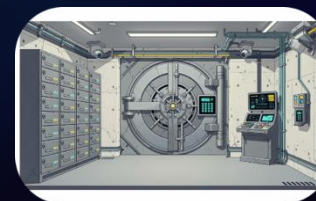
The VPC = The Fenced Property

The entire boundary line of the office park.
No outside roads can enter unless explicitly constructed by the developer.



Public Subnet = Front Lobby

Sits right by the entrance where delivery drivers, clients, and utility workers pull up to do daily business.



Private Subnet = Underground Vaults

Located deep beneath the compound. No windows, no public directories, and no doors leading to public streets.

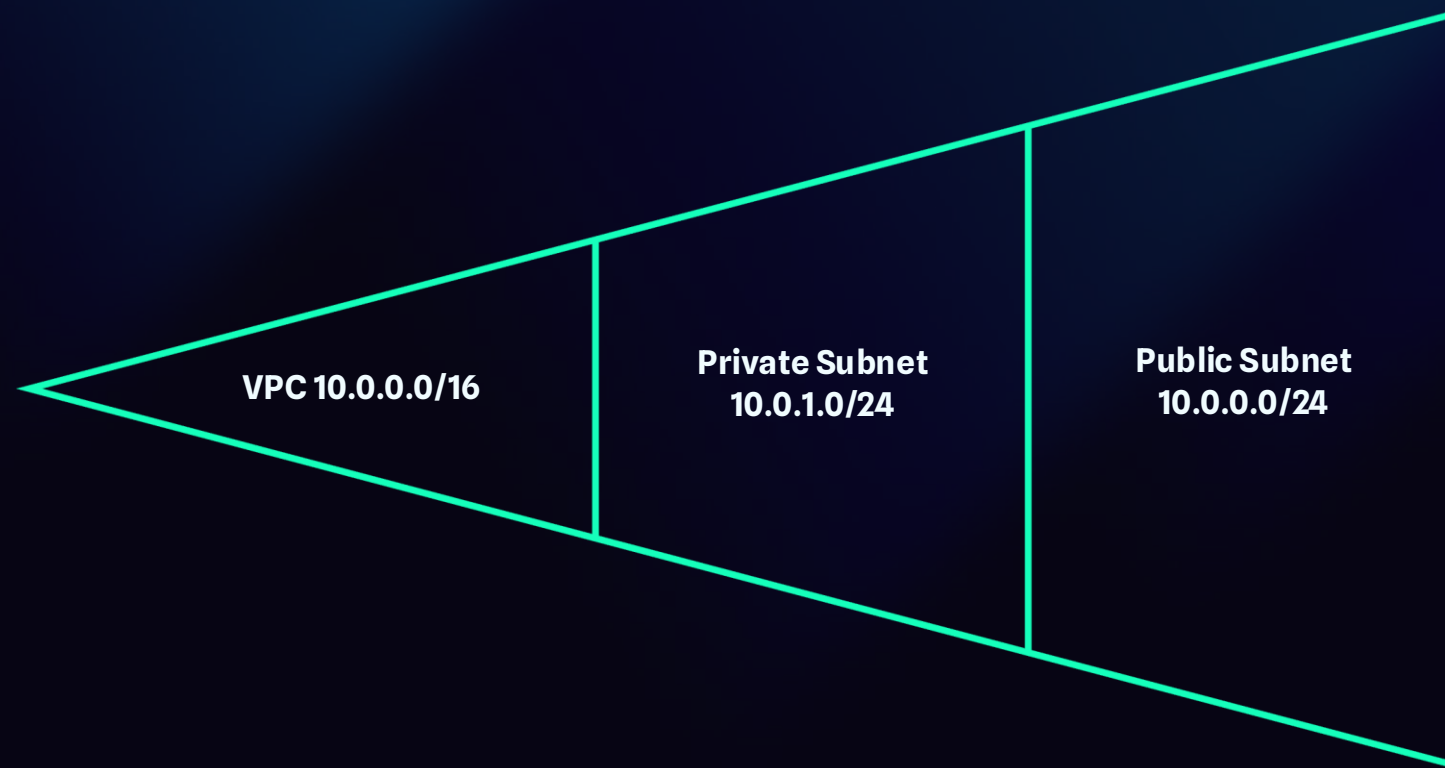
Chapter 2

CIDR Blocks & Subnets

Slicing up your IP address space into distinct neighborhoods

Demystifying CIDR Blocks

Before placing any servers, you must assign an address range to your digital land. **CIDR – Classless Inter-Domain Routing** – defines the range of unique IP addresses your VPC can use. Don't let the acronym scare you!



Think of a CIDR block as a **zip code** assigned to your neighborhood development. The **/16** notation means the first two sets of numbers are fixed for your neighborhood, giving you thousands of individual address plots to build on. Subnets let you divide that space into distinct, purposeful sections.

Public vs. Private Subnets

Public Subnet

Faces the open internet directly. This is where your consumer-facing web servers, load balancers, and NAT Gateways live. Traffic from the outside world can reach these resources — by design.

Example use: E-commerce storefront, login portal

Private Subnet

Completely disconnected from external view. This is where your databases, backend application servers, and sensitive records belong. No direct inbound path from the internet exists here.

Example use: Customer financial database, medical records

 **Think about it:** If a bank wants to protect customer records, should those database servers sit in a public subnet or a private subnet? Always private — and now you know exactly why.

The Residential Development Analogy

VPC = Total Land Boundary

The full property owned by the neighborhood construction corporation.

CIDR = Street Address Numbers

The total pool of available addresses allocated to the development project.

Public Subnet = Strip Mall

Commercial zone at the front – leasing office, visitor parking, open to all.

Private Subnet = Residential Cul-de-sacs

Hidden behind secondary security barriers – private homes, no public access.



Chapter 3

Internet Gateways & Route Tables

Traffic control – where your data is allowed to travel

The Internet Gateway (IGW)

Right now, our VPC is a fortress with no doors. To allow users to reach our web applications, we must attach an **Internet Gateway (IGW)** — the main public entrance gate connecting the VPC to the outside web.



The Main Gate

The IGW is the motorized checkpoint gate connecting your VPC's main access road to the public city highway. Remove it, and the entire park is cut off from external traffic.



One Per VPC

You can only attach one Internet Gateway per VPC — it's the single controlled entry and exit point for internet-bound traffic.



Not Enough Alone

Attaching an IGW doesn't automatically route traffic. You must also configure the Route Table to direct traffic toward it.

Route Tables: The Traffic Map

Once traffic passes through the Internet Gateway, how does it know which subnet to go to? That's where the **Route Table** comes in — a digital mapping spreadsheet listing exactly where network packets are allowed to travel.

The Road Map Analogy

Think of a Route Table as a giant directional map standing at the intersection of your neighborhood complex. It contains rules that dictate exactly where a packet of data can go.

For a subnet to be **truly public**, its Route Table must include this rule:

Destination: 0.0.0.0/0 → Target: IGW

Without this instruction, your server is completely unreachable from the outside world.

Route Table Rules Explained

Destination	Target	Meaning
10.0.0.0/16	local	Stay inside the VPC
0.0.0.0/0	IGW	Send to the internet (public)
0.0.0.0/0	NAT GW	Outbound only (private)

How Public Traffic Flows: Step by Step



Every packet traveling from the internet to your web server follows this exact path. The Route Table is the critical decision-maker — without a rule pointing to the IGW, traffic has nowhere to go and the server stays dark. **Route Tables are what make a subnet public or private.**

Chapter 4

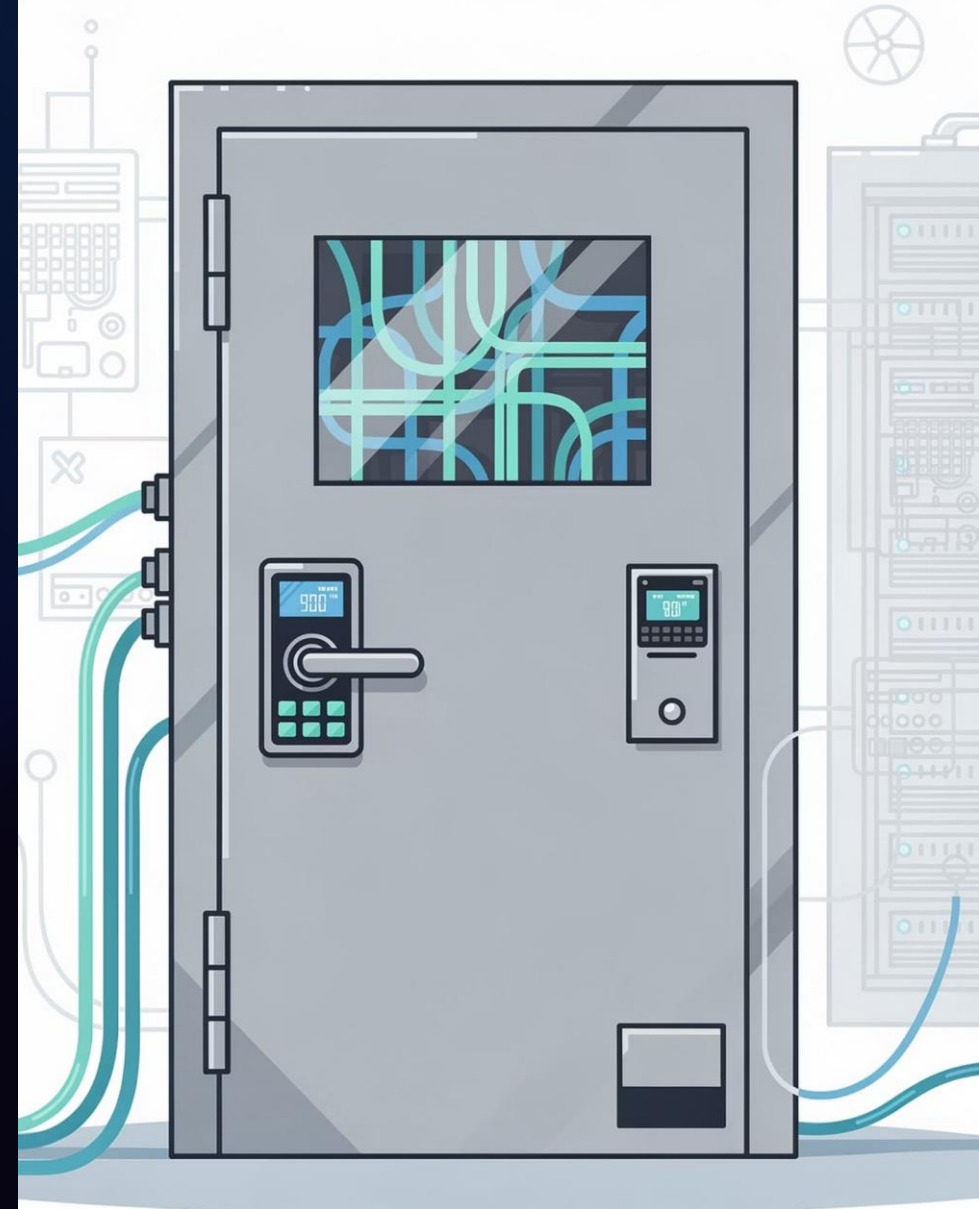
NAT Gateways & Bastion Hosts

Secure outbound paths and admin access — without opening the back door

The Problem with Private Subnets

Our backend database servers belong in the private subnet – they should never be directly reachable by random web traffic. But what happens when a database server needs to download a critical security patch? It needs a way to talk *out* to the internet without allowing the internet to talk *in*.

- This is one of the most common real-world challenges for junior cloud engineers. Knowing the solution makes you immediately valuable to an employer.



NAT Gateway: The One-Way Mirror

How It Works

A **Network Address Translation (NAT) Gateway** is deployed inside your Public Subnet. Private instances route their outbound traffic through it to reach the internet — but external parties cannot use that same path to initiate a connection back inside.

Think of it as a **secure outbound revolving door**. The private database server pushes a request out to grab updates. An external hacker cannot reverse-engineer that same path to reach the database.

The Mailroom Tube Analogy

Like a secure outbound shipping tube in the internal mailroom. Underground vault staff can drop orders into the tube to ship items to external vendors — but the tube's mechanics physically prevent any outside entity from crawling back up inside the vault.

-  Private instance sends request **out**
-  Internet responds to the NAT address
-  External parties cannot initiate **inbound** connections

Bastion Host: The Secure Admin Checkpoint

If an authorized administrator needs to log in and perform maintenance on a private database, they don't use the NAT Gateway. Instead, they connect through a **Bastion Host** — also called a **jump box**.

What It Is

A hardened, heavily locked-down server sitting in the Public Subnet. It serves as a security transition checkpoint — administrators authenticate here first, then access private backend resources from the inside.

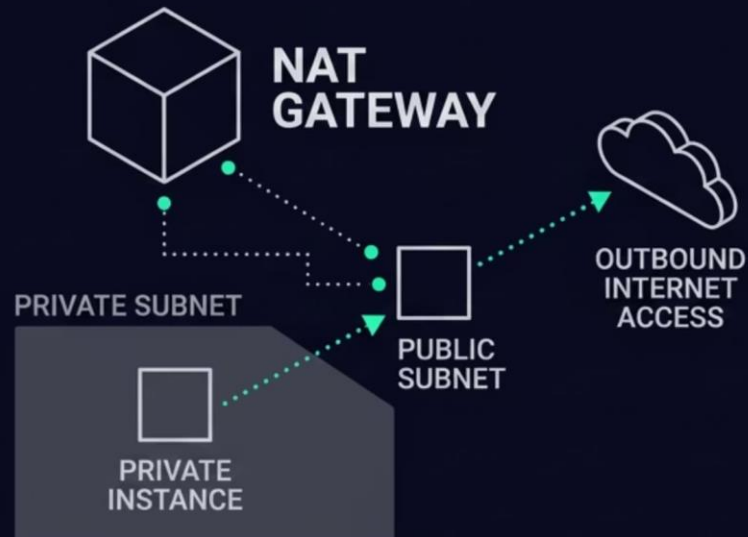
Why It Matters

Without a bastion host, admins would need to open the private subnet directly to the internet for management access — a critical security mistake. The bastion creates a controlled, auditable, single point of entry.

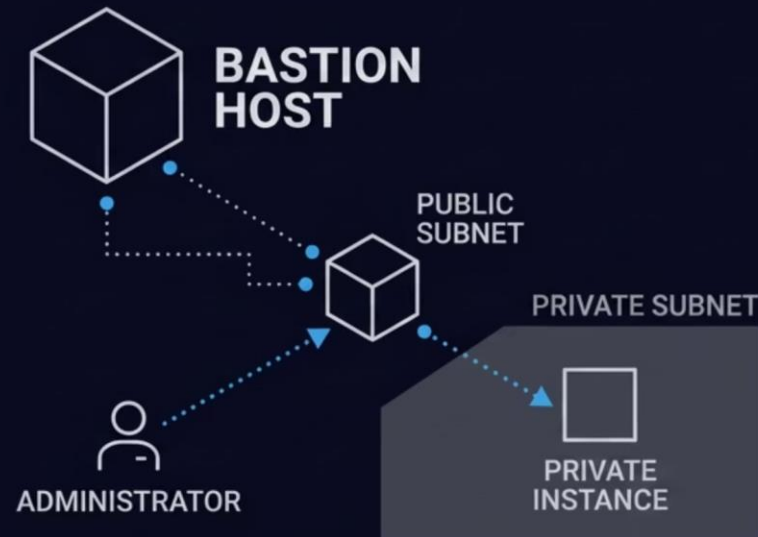
Career Connection

Setting up and hardening a bastion host is a real job task that appears regularly in cloud ops and sysadmin job descriptions. Know it cold.

NAT Gateway vs. Bastion Host: Side by Side



- **Purpose:** Outbound internet access
- **Direction:** Outbound only
- **Used by:** Servers/instances
- **Key Use:** Software updates
- **Lives:** Public subnet
- **Authentication:** No



- **Purpose:** Secure admin access
- **Direction:** Inbound SSH access
- **Used by:** Human administrators
- **Key Use:** Maintenance
- **Lives:** Public subnet
- **Authentication:** Strong (key pairs)

Chapter 5

Security Groups & NACLs

Two firewalls – one at your door, one at the neighborhood gate



AWS Provides Two Distinct Firewall Layers

To build a truly secure cloud environment, AWS gives you **two types of network firewalls** that work together. Understanding both — and their critical differences — is one of the most tested topics in AWS certifications and cloud job interviews.

Security Groups

Guard the individual **instance level** — like a personal bouncer at your server's front door. Stateful. Smart. Remembers who already came in.

Network ACLs (NACLs)

Guard the entire **subnet perimeter** — like a checkpoint at the neighborhood gate. Stateless. Checks every vehicle entering and leaving, no exceptions.

Security Groups: Stateful Protection

The Bouncer at Your Door

A Security Group operates directly at the virtual server (EC2 instance) level. It acts like a personal bouncer stationed at your specific executive office suite door.

Because Security Groups are **stateful**, they track connection state. If a visitor is verified and allowed inside, the bouncer remembers their face — and automatically lets them leave without checking their ID again.

Key Characteristics

- **Stateful** — return traffic is automatically allowed
- Supports **Allow rules only** — no explicit deny needed
- Applied at the **instance level**
- Can reference other Security Groups as source
- Default: deny all inbound, allow all outbound

📌 You can stack multiple Security Groups on one instance — giving you layered control over different traffic types.

NACLs: Stateless Subnet Guards

Key Characteristics

- **Stateless** — return traffic must be explicitly allowed
- Supports both **Allow AND Deny** rules
- Applied at the **subnet level**
- Rules are evaluated in **numbered order** (lowest first)
- Default NACL: allows all traffic; Custom NACL: denies all

❏ Because NACLs are stateless, forgetting to write an outbound rule is one of the most common beginner mistakes. Always write both inbound AND outbound rules!

The Perimeter Gate Guard

A NACL is like the security guard standing at the main iron gate of a private gated community. It checks **every** car entering or leaving the neighborhood — no exceptions.

Even if the guard lets a visitor drive in, that same visitor is **stopped and re-checked** against a separate outbound rules sheet when trying to leave. It has no memory of the earlier interaction.

Security Group vs. NACL: Full Comparison

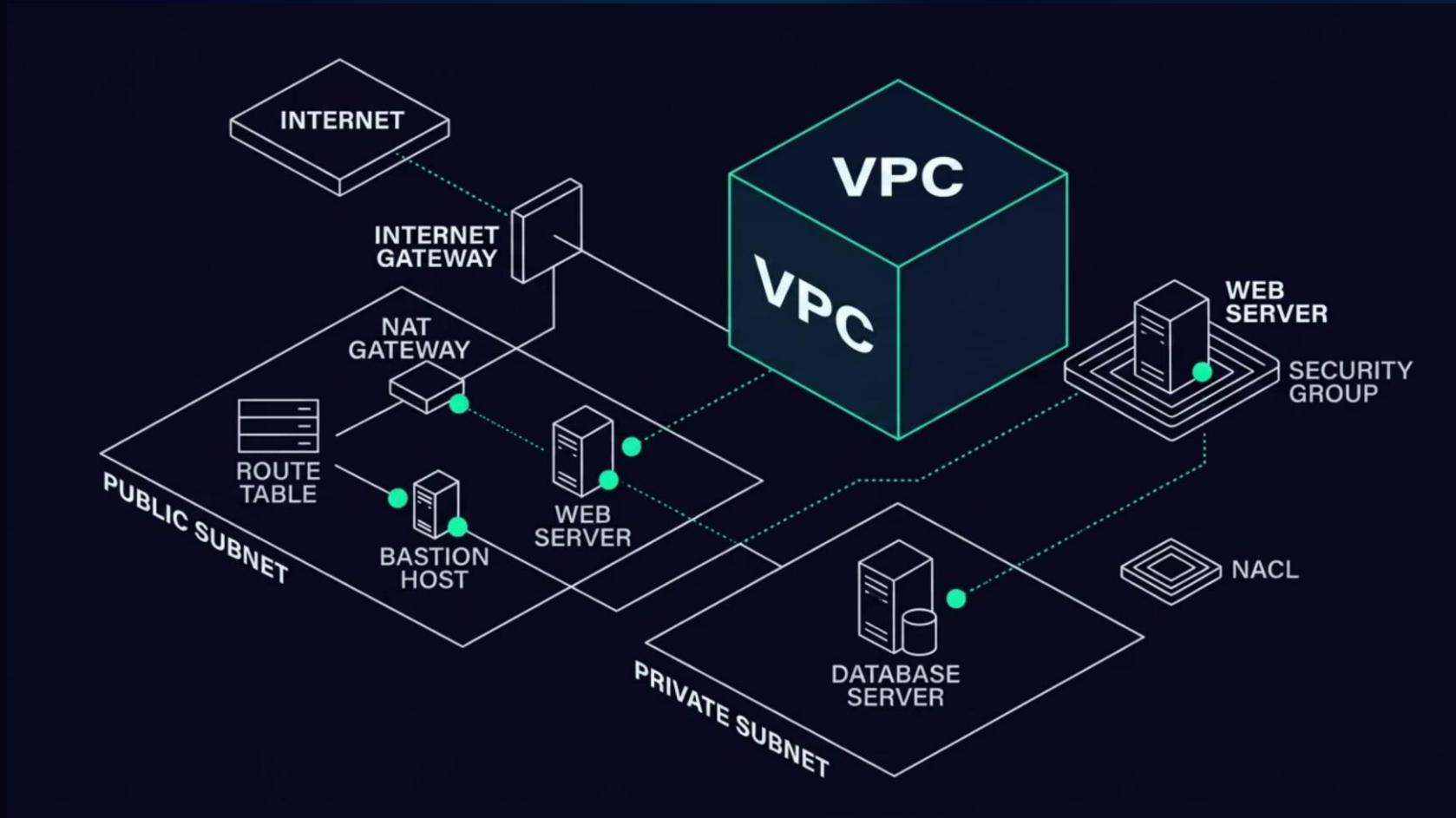
Feature	Security Group	Network ACL (NACL)
Operates at	Instance level (server door)	Subnet level (neighborhood gate)
State behavior	Stateful – return traffic auto-allowed	Stateless – return traffic must be explicit
Rule types	Allow rules only	Allow AND Deny rules
Rule evaluation	All rules evaluated together	Rules evaluated in number order
Default behavior	Deny all inbound; allow all outbound	Default NACL allows all traffic
Best used for	Specific port/IP control per server	Blocking broad IP ranges at subnet border

Traffic Order: Which Firewall Hits First?



When an inbound packet arrives at your subnet, it hits the **NACL first** — the neighborhood gate. If it passes that check, it then reaches the **Security Group** at the individual server door. Both must allow the traffic for it to reach your application. This two-layer approach is your defense-in-depth strategy.

The Full VPC Architecture at a Glance



Every component we've covered today works as part of this layered defense architecture. The IGW is the only front door. The Route Table directs traffic. NAT handles outbound updates. The NACL guards the subnet gate. Security Groups guard each server door.

Chapter 6

In-Class Activity

The Cloud Packet Flow Drawing Challenge

Activity: The Cloud Packet Flow Drawing Challenge

Format

Think-Pair-Share &
Collaborative
Architectural
Whiteboarding

Time

20 minutes total —
5 min solo review,
10 min pair
drawing, 5 min
class debrief

Goal

Map a live packet's
journey through
every VPC
component — from
internet user to
database server



Your Scenario: Junior Cloud Support Technician

"You are working as a Junior Cloud Support Technician for an e-commerce platform. The senior engineering team has launched a web server in a public subnet and a MySQL database server in a private subnet inside a custom VPC. A user clicks a button on their browser to view an item. This sends an inbound HTTP packet on Port 80 from their public IP across the internet to the web server. The web server processes the request and must query the database in the private subnet."

Working in pairs: Map the packet's journey step-by-step through every networking and security component — from the external user all the way to the internal database. Then identify the exact rules that prevent the user from connecting directly to the database while allowing the web server to reach it safely.

Activity Evaluation Rubric

Performance Criterion	Excellent (5 pts)	Developing (3 pts)	Unsatisfactory (1 pt)
Ingress Path Mapping	Correctly traces IGW → NACL → Security Group in order	Traces path but confuses NACL and Security Group sequence	Misses multiple core components; traffic skips boundaries
Component Rule Definition	Web SG allows port 80; DB SG restricts to web server's internal IP	Identifies firewalls must be open but leaves DB wide open (0.0.0.0/0)	Fails to specify any port numbers or IP boundary settings
Isolation Verification	Accurately identifies private subnet has no IGW route, explains how Route Table enforces isolation	Knows DB is private but can't explain how Route Table enforces it	Suggests adding an IGW directly to the private subnet

Debrief: Three Key Questions

1 Which firewall does an inbound packet hit first — the NACL or the Security Group?

Answer: The NACL first — it guards the entire subnet perimeter gate before the Security Group ever sees the packet at the individual server door.

2 If the NACL allows port 80 traffic in, do we need a manual outbound rule to send the response back?

Answer: Yes! NACLs are stateless — they don't remember who came in. You must explicitly write an outbound rule to allow the server's response back out to the customer.

3 What source IP should the database Security Group allow inbound connections from?

Answer: Only the private IP address — or the Security Group ID — of the frontend web server. All other public traffic sources must be completely blocked.

Chapter 7

Vocational Case Study

Architectural Failure Analysis: The Exposed Patient Directory

Case Study: MedVanguard Solutions

A regional healthcare analytics provider migrated their electronic medical records application to AWS. The technical layout — designed by a junior contractor — placed the consumer portal in a public subnet and a backend EC2 database instance containing the **unencrypted Protected Health Information (PHI) of over 45,000 patients** in what was intended to be a private tier.

- ⊗ What happened next is one of the most important lessons you'll take from this class — because this exact mistake is made on real systems every year.



What Went Wrong: The Two Critical Errors

Error #1: Single Shared Route Table

The contractor created a singular master Route Table containing a rule: **0.0.0.0/0** → **IGW**, and bound *all* subnets to it — including the database subnet. Because the Route Table pointed to the Internet Gateway, the database subnet was automatically classified as a **public subnet**, fully exposed to the internet.

Error #2: Unrestricted Security Group

During initial testing, the database server's Security Group inbound rule was set to **0.0.0.0/0** — allowing all traffic from anywhere — to bypass credential errors encountered during setup. This configuration was never corrected before launch, removing the final layer of instance-level protection entirely.

The Fallout: Real Consequences

45K

Patient Records Exposed

Unencrypted Protected Health Information (PHI) discoverable from the open internet

\$120K

Forensic Audit Penalties

Mandatory external cybersecurity audit costs levied immediately after the violation finding

3306

Database Port Exposed

MySQL's default port — discoverable by any routine internet-wide port scan

The regulatory compliance audit flagged the configuration as a **federal security mandate violation**. MedVanguard received an immediate administrative stop-work order and was required to rebuild their entire cloud networking architecture from scratch before resuming operations.

Debate: What Would You Have Done Differently?

These questions are designed for pair or small group discussion. Think like a cloud security professional before reading the answers.

1

The Route Table Question

Which specific configuration oversight converted the intended private database subnet into a publicly accessible zone? How could it have been caught before launch?

2

The Least Privilege Question

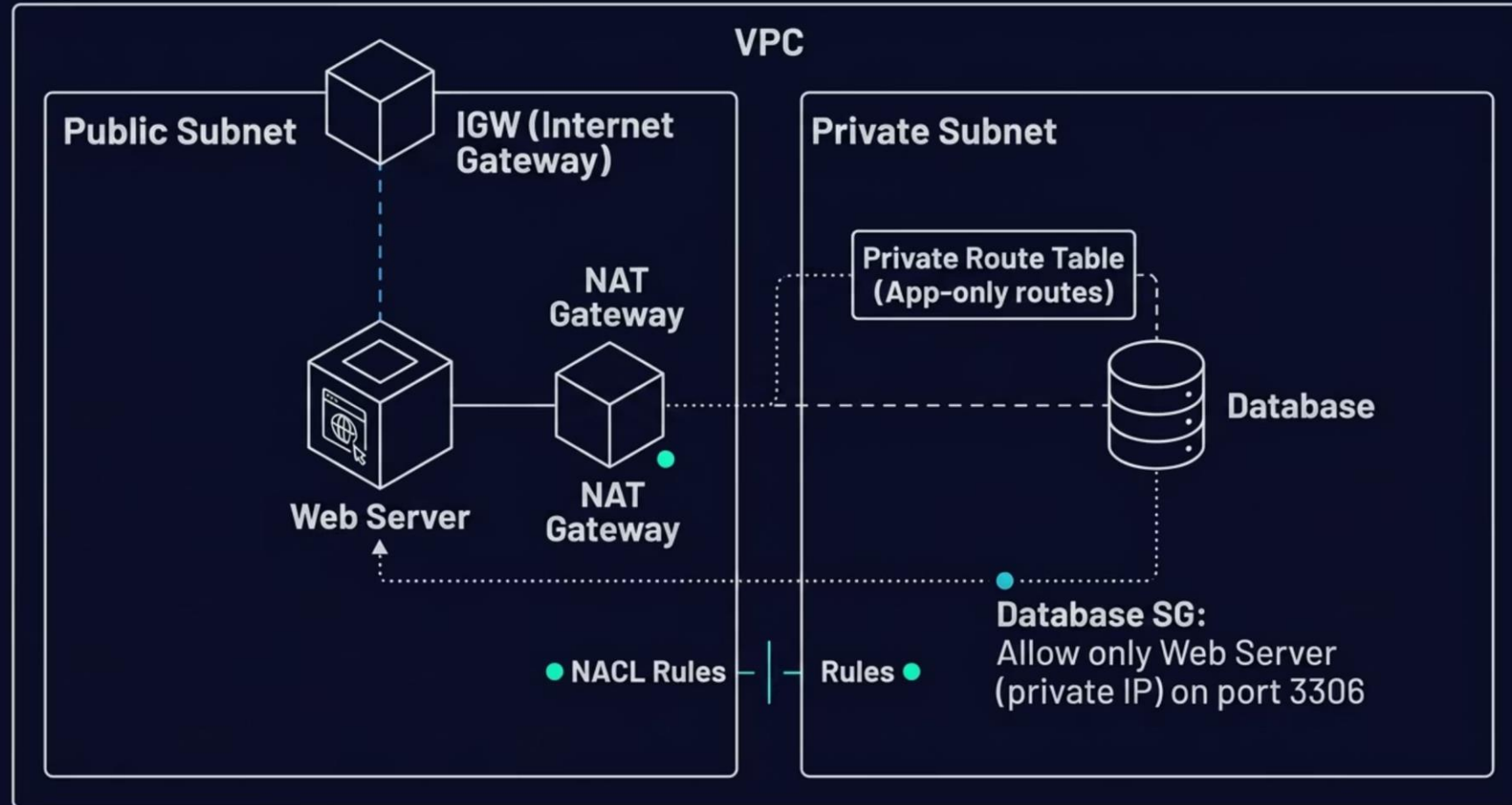
How could applying the Principle of Least Privilege within the Security Group rules have prevented exposure — even with the incorrect Route Table in place?

3

The Remediation Question

You're brought in as the new Lead Cloud Infrastructure Analyst. Outline the exact architectural changes you'd enforce to guarantee the database never experiences public exposure again.

The Correct Architecture: What It Should Have Looked Like



Four architectural fixes. One correctly isolated database. Zero public exposure. This is the standard two-tier architecture pattern that employers expect junior cloud engineers to design from memory.

Chapter 8

Full VPC Analogy Review

Putting it all together with the Office Park component matrix

The Complete Office Park Component Matrix

VPC Component	Office Park Equivalent
VPC	The entire fenced-in corporate property – no external roads enter unless built by the developer
CIDR Block	The master land registry map with exact lot numbers and street coordinates
Public Subnet	Front lobby reception building – delivery drivers and clients pull up here daily
Private Subnet	Underground research vaults – no windows, no public directory, no exterior doors
Internet Gateway	Motorized vehicle checkpoint gate connecting the park's main road to the public city highway
Route Table	Graphical map directory at all driving intersections – tells vehicles which way to go
NAT Gateway	Secure outbound shipping tube – vault staff send orders out; nothing crawls back up inside
Security Group	Guard at a specific executive suite door – stateful, remembers faces, lets vetted workers leave freely
NACL	Iron turnstile at elevator bank entrance – screens everyone entering AND exiting against a strict master list

Chapter 9

Key Takeaways

What you now know that employers are looking for

The Five Things You Must Know Cold



VPC = Your Isolated Private Network

Your virtual land inside AWS — you control the layout, IPs, routing, and security entirely through software.



CIDR + Subnets = Divided Neighborhoods

CIDR defines your total address space. Subnets divide it into public-facing and privately isolated zones.



IGW + Route Table = Public Access

An IGW opens the door. A Route Table rule (0.0.0.0/0 → IGW) is what actually makes a subnet public.



NAT Gateway = Safe Outbound Updates

Private instances update through the NAT Gateway — outbound only, no inbound access possible from outside.



NACLs + Security Groups = Layered Defense

NACLs guard subnet borders (stateless). Security Groups guard instance doors (stateful). Use both every time.

The Most Common Mistakes on the Job

→ **Forgetting the NACL Outbound Rule**

Because NACLs are stateless, a missing outbound rule will silently break your application – even though the inbound rule looks correct. Always write both directions explicitly.

→ **Binding All Subnets to One Route Table**

A single shared Route Table with an IGW route makes every subnet public – even the ones you intended to be private. Always create a dedicated Private Route Table.

→ **Using 0.0.0.0/0 on a Database Security Group**

The MedVanguard case study is not fiction – this happens constantly. Lock your database Security Group to the specific web server IP or Security Group ID only.

→ **Confusing NAT Gateway and Bastion Host Roles**

NAT Gateways are for server-to-internet outbound traffic. Bastion Hosts are for human administrator inbound access. These are two completely different tools with different purposes.

How This Maps to Real Job Roles



Cloud Engineer

Designs and deploys VPCs, configures subnets, attaches gateways, and builds Route Tables for production workloads. VPC is the foundational skill for this role.



Network Operations Specialist

Monitors traffic flows, troubleshoots routing issues, and maintains Security Group and NACL rule sets. This lesson is their daily work.

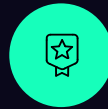


Cloud Security Analyst

Audits VPC configurations for compliance violations, identifies exposed resources, and remediates Security Group misconfigurations — exactly like the MedVanguard scenario.



Certifications This Lesson Prepares You For



AWS Cloud Practitioner

VPC, subnets, Security Groups, and NACLs appear directly in the foundational exam. This lesson covers the core concepts you'll need.



AWS Solutions Architect - Associate

Route Tables, NAT Gateways, Bastion Hosts, and multi-tier VPC architecture design are heavily tested at this level.



CompTIA Cloud+

Vendor-neutral cloud networking concepts including subnetting, routing, and firewall layering align directly with today's material.

Quick Knowledge Check

Test yourself before the next class. If you can answer all five confidently, you've owned today's material.

1

What makes a subnet "public"?

A Route Table with a rule pointing 0.0.0.0/0 to an Internet Gateway attached to the VPC.

2

Why do NACLs require explicit outbound rules?

Because NACLs are stateless – they have no memory of inbound connections and treat every packet independently.

3

What is the difference between a NAT Gateway and a Bastion Host?

NAT Gateway enables outbound internet access for private servers. Bastion Host provides secure admin login access to private resources.

4

Which firewall layer is checked first – NACL or Security Group?

NACL – it operates at the subnet boundary and is evaluated before the Security Group at the instance door.

5

What two mistakes caused the MedVanguard breach?

A single shared Route Table with an IGW route (making the DB subnet public) and a 0.0.0.0/0 inbound Security Group rule on the database instance.

Looking Ahead: What's Next

Your Homework Before Next Class

- Review today's vocabulary terms – write each one and its analogy in your own words
- Sketch a two-tier VPC diagram from memory: public subnet with IGW and NAT Gateway, private subnet with database
- Read the AWS documentation summary on Security Groups vs. NACLs
- Think about which job role from Slide 45 aligns most with your career goals – we'll discuss next class

Coming Up in Day 9

We'll build on today's VPC foundation and dive into:

- **AWS IAM** – Identity and Access Management for controlling who can do what inside your cloud
- **Roles vs. Policies** – how permissions are structured and attached to users and services
- **The Principle of Least Privilege** – the security philosophy that underpins everything we've built today

Today's material is the networking foundation. Day 9 adds the identity layer on top. Together they form the full security posture of a professional cloud environment.

You've Got This — And You've Already Proven It

Today's material is exactly what cloud employers test in technical interviews. If you can explain the difference between a Security Group and a NACL, design a two-tier VPC architecture, and explain why the MedVanguard breach happened — you are already ahead of most entry-level candidates walking into those interviews.

"The professionals who truly understand cloud networking — not just the surface terminology, but the *why* behind every design decision — are the ones who get hired, get promoted, and get trusted with the most important systems. That's exactly what you're building right now."

Great work today. See you in Day 9. 🚀

